

Title

Comparative effectiveness of digital smoking-cessation coaching with varying levels of synchronous therapeutic support: protocol for a three-arm randomized controlled trial

Abstract

Mobile-application-based smoking cessation interventions are increasingly deployed in clinical practice, yet evidence regarding optimal implementation--particularly whether augmenting app-based self-guided modules with human therapeutic contact enhances cessation outcomes--remains limited. This protocol describes a three-arm randomized trial enrolling 420 adults (140 participants per arm) actively seeking smoking cessation support. Participants will use a cognitive-behavioral mobile application over eight weeks, randomly assigned to: (1) autonomous app-only self-guided use; (2) app paired with eight weekly group-based cognitive-behavioral therapy sessions delivered via secure videoconference; or (3) app paired with four biweekly group cognitive-behavioral sessions via videoconference. Primary outcome is validated smoking abstinence (exhaled carbon monoxide ≤ 6 ppm) measured at end-of-treatment. Secondary outcomes assess sustained abstinence through 12 months post-treatment, quit-attempt patterns, treatment satisfaction, and healthcare utilization. This trial will determine whether supplementing digital cessation tools with graduated human therapeutic engagement improves smoking abstinence outcomes and will guide optimization of remote smoking-cessation service delivery.